sheet (3) section (D)

Reduction Formulas

$$\boxed{I_n = \int \sin^n x \, dx = \frac{-1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} I_{n-2}} \quad n \geq 2$$

proof

$I_n = \int \sin^{n-1} x \, \sin x \, dx$ ← by parts

$u = \sin^{n-1} x$, $dv = \sin x \, dx$

$du = (n-1) \sin^{n-2} x \cos x$, $v = -\cos x$

$I_n = uv - \int v \, du$

$I_n = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx$

$I_n = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, (1 - \sin^2 x) \, dx$

$I_n = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x - \sin^n x \, dx$

$I_n = -\cos x \sin^{n-1} x + (n-1) \left[\int \sin^{n-2} x \, dx - \int \sin^n x \, dx \right]$

$I_n = -\cos x \sin^{n-1} x + (n-1) \left[I_{n-2} - I_n \right]$

$I_n = -\cos x \sin^{n-1} x + (n-1) I_{n-2} - (n-1) I_n$

$I_n + (n-1) I_n = -\cos x \sin^{n-1} x + (n-1) I_{n-2}$

$\cancel{I_n} + n I_n - \cancel{I_n} = -\cos x \sin^{n-1} x + (n-1) I_{n-2}$

$\therefore I_n = \frac{-1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} I_{n-2}$ *

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$$\boxed{I_n = \int \cos^n x \, dx = \frac{1}{n} \sin x \cos^{n-1} x + \frac{n-1}{n} I_{n-2}} \quad n \geq 2$$

Proof

$I_n = \int \cos^{n-1} x \cos x \, dx$ ← by parts | $u = \cos^{n-1} x$, $dv = \cos x \, dx$

$I_n = uv - \int v \, du$ | $u = -(n-1)\cos^{n-2} x \sin x$, $v = \sin x$

$$\therefore I_n = \sin x \cos^{n-1} x + (n-1) \int \cos^{n-2} x \sin^2 x \, dx$$

$$I_n = \sin x \cos^{n-1} x + (n-1) \int \cos^{n-2} x (1 - \cos^2 x) \, dx$$

$$I_n = \sin x \cos^{n-1} x + (n-1) \int \cos^{n-2} x - \cos^n x \, dx$$

$$I_n = \sin x \cos^{n-1} x + (n-1) \left[\int \cos^{n-2} x \, dx - \int \cos^n x \, dx \right]$$

$$I_n = \sin x \cos^{n-1} x + (n-1) \left[I_{n-2} - I_n \right]$$

$$I_n = \sin x \cos^{n-1} x + (n-1) I_{n-2} - (n-1) I_n$$

$$I_n + (n-1) I_n = \sin x \cos^{n-1} x + (n-1) I_{n-2}$$

$$\cancel{I_n} + n I_n - \cancel{I_n} = \sin x \cos^{n-1} x + (n-1) I_{n-2}$$

$$\therefore I_n = \frac{1}{n} \sin x \cos^{n-1} x + \frac{n-1}{n} I_{n-2} \quad *$$

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$$\boxed{I_n = \int \sec^n x \, dx = \frac{1}{n-1} \tan x \sec^{n-2} x + \frac{n-2}{n-1} I_{n-2}} \qquad n \geq 2$$

Proof

$I_n = \int \sec^{n-2} x \sec^2 x \, dx$ ← by parts

$u = (\sec x)^{n-2}$, $dv = \sec^2 x \, dx$

$du = (n-2) \sec^{n-2} x \tan x$, $v = \tan x$

$I_n = uv - \int v \, du$

$$I_n = \tan x \sec^{n-2} x - (n-2) \int \sec^{n-2} x \tan^2 x \, dx$$

$$I_n = \tan x \sec^{n-2} x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) \, dx$$

$$I_n = \tan x \sec^{n-2} x - (n-2) \int \sec^n x - \sec^{n-2} x \, dx$$

$$I_n = \tan x \sec^{n-2} x - (n-2) \left[\int \sec^n x \, dx - \int \sec^{n-2} x \, dx \right]$$

$$I_n = \tan x \sec^{n-2} x - (n-2) \left[I_n - I_{n-2} \right]$$

$$I_n = \tan x \sec^{n-2} x - (n-2) I_n + (n-2) I_{n-2}$$

$$I_n + (n-2) I_n = \tan x \sec^{n-2} x + (n-2) I_{n-2}$$

$$(1+n-2) I_n = \tan x \sec^{n-2} x + (n-2) I_{n-2}$$

$$\therefore I_n = \frac{1}{n-1} \tan x \sec^{n-2} x + \frac{n-2}{n-1} I_{n-2} \quad *$$

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$$\boxed{I_n = \int \tan^n x \, dx = \frac{1}{n-1} \tan^{n-1} x - I_{n-2}} \quad n \geq 2$$

Proof

$$I_n = \int \tan^2 x \, \tan^{n-2} x \, dx$$

$$I_n = \int (\sec^2 x - 1) \tan^{n-2} x \, dx$$

$$I_n = \int \sec^2 x \tan^{n-2} x - \tan^{n-2} x \, dx$$

$$I_n = \int \sec^2 x \, (\tan x)^{n-2} dx - \int \tan^{n-2} x \, dx$$

$$I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2} \quad *$$

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(41) Prove that $\int \sec^n x\,dx = I_n = \frac{1}{n-1}\tan x \sec^{n-2} x + \frac{n-2}{n-1} I_{n-2}$

& hence find $\int \sec^5 x\,dx$

Sol^n

"you prove the formula as given before, then use it to solve the problem"

$$\int \sec^5 x\,dx = I_5$$

$n=5$

$$\longrightarrow I_5 = \frac{1}{4}\tan x \sec^3 x + \frac{3}{4} I_3$$

$n=3$

$$\longrightarrow I_3 = \frac{1}{2}\tan x \sec x + \frac{1}{2} I_1 \rightarrow \int \sec^1 x\,dx$$

$$= \frac{1}{2}\tan x \sec x + \frac{1}{2}\ln|\sec x + \tan x| + c$$

$$\therefore I_5 = \frac{1}{4}\tan x \sec^3 x + \frac{3}{4}\left[\frac{1}{2}\tan x \sec x + \frac{1}{2}\ln|\sec x + \tan x|\right] + c$$

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(41) Prove that $\int \tan^n x \, dx = I_n = \frac{1}{n-1} \tan^{n-1} x - I_{n-2}$

& hence find $\int \tan^7 x \cot^2 x \, dx$

Sol^n:

"you prove the formula, then use it in solving the problem"

$$\int \tan^7 x \cot^2 x = \int \tan^5 x \, dx = I_5$$

$n=5$

$\longrightarrow I_5 = \frac{1}{4} \tan^4 x - I_3$

$n=3$

$\longrightarrow I_3 = \frac{1}{2} \tan^2 x - I_1 \longrightarrow \int \tan^1 x \, dx$

$$I_3 = \frac{1}{2} \tan^2 x - \ln|\sec x| + C$$

$$\therefore I_5 = \frac{1}{4} \tan^4 x - \left[\tfrac{1}{2} \tan^2 x - \ln|\sec x| \right] + C$$

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(39) Prove that $\int \sin^n x\, dx = I_n = \frac{-1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} I_{n-2}$

& hence find $\int \frac{\sin^4 x}{\operatorname{cosec}^2 x}\, dx$

Sol^n

"you prove the formula as given before then you use it to solve the problem"

$$\int \frac{\sin^4 x}{\operatorname{cosec}^2 x}\, dx = \int \sin^6 x\, dx = I_6$$

$n=6$

$$\rightarrow I_6 = \frac{-1}{6} \sin^5 x \cos x + \frac{5}{6} I_4$$

$n=4$

$$\rightarrow I_4 = \frac{-1}{4} \sin^3 x \cos x + \frac{3}{4} I_2$$

$n=2$

$$\rightarrow I_2 = \frac{-1}{2} \sin x \cos x + \frac{1}{2} I_0 \rightarrow \int \underbrace{\sin^0 x}_{1}\, dx$$

$$I_2 = -\frac{1}{2} \sin x \cos x + \frac{1}{2} x + C$$

$$\therefore I_6 = -\frac{1}{6} \sin^5 x \cos x + \frac{5}{6} \left\{ \frac{-1}{4} \sin^3 x \cos x + \frac{3}{4} \left[\frac{-1}{2} \sin x \cos x + \frac{1}{2} x \right] \right\} + C$$

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